Mathematical Preparation for Ph.D. Studies in Public Policy and Public Affairs

Summer 2026 — Asynchronous Online

Professors:	Alberto Ortega (alorte@iu.edu)	**Format:**	Asynchronous
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		Location:	SPEA 270

Course Description

This course is a two-week review of mathematics intended to prepare incoming Ph.D. students for subsequent courses at the O'Neill School. It is delivered asynchronously and organized into ten one-day modules. Modules 1–6 review algebra, functions, calculus, optimization, integration, and matrix algebra — with economic applications (supply and demand, production and costs, profit maximization, consumer choice) woven throughout. Modules 7–10 continue into **probability and statistics**.

For some of you, this material will be a review. For others, much of it will be new. The emphasis is not on proving theorems but rather on learning how to use mathematical concepts. Each module includes a problem set so you can work through problems independently. If you still don't grasp a concept after working through the assigned problems, it is a good idea to work through some additional ones (see the "Texts with additional problems" list below).

How the Course Works

You are highly, highly, highly encouraged to **meet with your incoming classmates and work through this course together**. **SPEA 270** is reserved for you during the camp (August 3–14). All course materials — notes, slides, problem sets, and solutions — are posted on the **course website**: `https://oneill-quant.github.io/`.

Each module is designed to take **one day**, following the schedule on the next page. We suggest the following format for each day:

- **10:00 a.m.** Meet in SPEA 270 and work through the module's lecture notes together.
- **Midday.** Attempt the module's problem set on your own. Give every problem an honest attempt before looking anything up.
- **2:00 p.m.** Reconvene after lunch to compare answers, discuss the problem set, and work through anything that didn't click — or do coding assignments.
- **Evening.** Review the day's material and read ahead for the next day's module.

- **Following day.** Turn in your **handwritten** problem set: scan or photograph it and upload it to **Canvas**. If a problem set involves code, upload your scripts as well.

Work through the modules **in order** — each one builds on the previous. If you have questions at any point, email us; we are happy to set up a meeting. Whatever happens, complete all modules before the start of the fall semester.

Assignments

There are **ten homework assignments** (one problem set per module) and a **final exam**.

- **Homework (ungraded).** Each module's problem set is turned in on **Canvas** the following day but is **not graded**. The problem sets are where the learning happens — give them your full effort.
- **Final exam (graded).** The final is taken **in person**, without the aid of AI tools, and **is graded**. More details to come on the time and place of the final exam.

The final exam grade will **not** count toward your Ph.D. GPA or matriculation. It will, however, allow the faculty to see where you are mathematically — and how seriously you took the math camp and the start of the program.

Textbooks

Camp readings come from the two texts below: (MS) for the mathematics modules and (BH) for probability and statistics. The Moore & Siegel text (MS) is posted on **Canvas**, where you can download it. The other texts listed further down are optional references — helpful if you want another explanation or extra practice — but you do not need to acquire them.

- (MS) Will H. Moore and David A. Siegel. *A Mathematics Course for Political and Social Research.* Princeton University Press, 2013.
- (BH) Bruce E. Hansen. *Probability and Statistics for Economists.* Princeton University Press, 2021.
- Note that some material covered in this course is not covered in these texts.

<u>Other useful texts (prior editions of these texts are fine):</u>

- (SH) Knut Sydsæter, Peter Hammond, and Anne Strom. *Essential Mathematics for Economic Analysis*, Prentice Hall, 4th edition, 2012 (5th Edition is also fine)
- Jeff Gill. *Essential Mathematics for Political and Social Research*. Cambridge, 2006.
- Malcolm Pemberton and Nicholas Rau. *Mathematics for Economists: An Introductory Textbook.*

<u>More advanced texts:</u>

- Carl P. Simon and Lawrence Blume. *Mathematics for Economists*, W. W. Norton & Company, Inc, 1994
- Alpha C. Chiang and Kevin Wainwright. *Fundamental Methods of Mathematical Economics*. McGraw-Hill Companies, 4th edition, 2005

<u>Texts with additional problems:</u>

- Schaum's Outline, Introduction to Mathematical Economics
- Schaum's Outline, Outline of Statistics and Econometrics, Second Edition

Modules 1–6: Mathematics

Module	Date	Topics	MS	SH	Due
1. Algebra & Functions	Aug. 3	– Algebraic operations – Systems of equations – Basics of functions – Linear, power, exponential & log functions – Composite functions – Application: supply & demand	Chapters 2–3	Chapters 1–5	Problem Set 1
2. Limits, Continuity & Derivatives	Aug. 4	– Limits and continuity – The derivative: definition and interpretation – Rules of differentiation – Implicit differentiation – L'Hôpital's rule – Higher-order derivatives, graphing, local extrema & inflection points	Chapters 4–6 8.1–8.3	Chapters 6–7 8.1–8.3	Problem Set 2
3. Optimization & the Firm	Aug. 5	– Convexity & concavity – Unconstrained optimization – Extreme value theorem – Applications: production, costs, and profit maximization	Chapter 8 16.1	8.4–8.7	Problem Set 3
4. Integration	Aug. 6	– Antiderivatives – Rules of integration – Definite integrals – Integration by parts – Application: consumer & producer surplus	Chapter 7	Chapter 9	Problem Set 4
5. Matrix Algebra	Aug. 7	– Vectors and vector operations – Matrices and matrix operations – Transpose; vector spaces – Solving linear systems – Non-singularity, rank, determinants – Matrix inverse and Cramer's rule	Chapters 12–13	Chapters 15–16	Problem Set 5
6. Multivariable & Constrained Optimization	Aug. 10	– Functions of several variables – Partial derivatives – Multivariate unconstrained optimization – Necessary & sufficient conditions – Constrained optimization and the Lagrangian – Application: utility maximization	5.3.4 Chapters 15–16	Chapters 11, 13–14	Problem Set 6

Problem Set 2 also includes two exercises from Gill (5.8 and 6.9); they are restated in full on the problem set, so you do not need the book.

Modules 7–10: Probability & Statistics

Module	Date	Topics (tentative)	BH	Due
7. Foundations of Probability	Aug. 11	– Sample spaces & events – Probability axioms – Conditional probability & independence – Bayes' rule	Chapter 1	Problem Set 7
8. Random Variables & Distributions	Aug. 12	– Discrete & continuous random variables – Distributions & densities – Expectation & variance – Covariance & correlation	Chapter 2	Problem Set 8
9. Statistics & Statistical Inference	Aug. 13	– Random sampling – The law of large numbers & the CLT – Estimation & statistical inference	Chapter 3	Problem Set 9
10. Statistics & Statistical Inference, continued	Aug. 14	– Chapter 3, continued – Review and catch-up – Prep for the final exam	Chapter 3	Problem Set 10

Modules 7–10 continue your preparation into probability and statistics ahead of the first-year methods sequence. Readings are selected sections of the opening chapters of (BH); slides and problem sets will be posted as camp approaches.